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VANDALISM IN THE PIT AND ENGINEERING STEREOTYPES

MADDY WHITE
SECTION EDITOR - ARTS & CULTURE

Pictured: *The vandalism on LGBTQase's mural in the pit occured in November of this school year.*

Less than a year ago, a group that I'm a part of got together and painted a beautiful mural in the Atrium. Every time I walk by it, I feel proud to see us represented with our own space as a part of the engineering community.

Last semester, on Saturday November 23rd, I received word that the mural had been vandalized. I asked one question in response: 'Was the vandalism derogatory?'.

If the group was one of the technical or sports teams, I doubt I would have had to ask that question. But in fact, the group I am talking about is LGBTQase (Lesbians, Gays, Bisexuals, Trans People, Queers & Allies in Science and Engineering). And sadly, that question did need to be asked - because the answer was yes.

We don't know who did it. As I said, we first heard about the vandalism on Saturday. When we came in the next day to paint over it, we learned that the Atrium was rented out externally that weekend, so the perpetrator could have been anyone, not necessarily an engineering student.

To be honest, the possibility that it had been done by someone from outside of the community didn't actually occur to me until much later. I assumed that it was a student, most likely one in engineering, because engineers aren't often known for their sensitivity to queer issues. That's a stereotype of course, one that is not true to every engineer. But the stereotype exists because of things that have happened in the past, and it is something that belongs to us all now.

I think that the stereotypes often aren't recognized within the engineering community because it's so isolated. We all tend to hang out amongst ourselves, which means that we're often not confronted with the negative stereotypes about engineers until we have to step outside the community. I have walked into bars with an engineering jacket on, and had people look at me differently. I can't say it's a fun experience to know that people are assuming things about you that you know aren't true. Stereotypes are easy to ignore until they start affecting you personally.

This isolated quality of the community also makes it difficult to see where the stereotypes come from. It's hard to think of your friends, or even yourself, as being someone who perpetuates these stereotypes, because you know that you and your friends are 'good' people. So from your point of view, something

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EDITOR-IN-CHIEF
Ashkan Parcham-Kashani

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Vismit Joshi
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Vismit Joshi
Tracy Zahradnik
Meghan Lele
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Dana Umbrio
Israel Monzon Gonzalez
Nikita Dawe

PHOTOGRAPHER
William Shenghua Ye

WEB DEVELOPERS
Maged Ahmed
Mark Ye
Meghan Lele
Nyantara Prem
Edgar Wang

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THE CANNON
10 King's College Road
Sandford Fleming Building
Room B740
Toronto, ON
M5S 3G4

cannon@skule.ca
cannon.skule.ca

Vandalism in the Pit, continued from page 1.

your friend does is easily classified as being a harmless joke, since you know that in all likelihood they weren't intending real harm (even if that was the end result).

I am sure that there will be people who read this article who will say that LGBTQase and I took this incident too seriously. But that is exactly the virus at the root of the disease, this idea that anything is okay if it was no big deal to you, and the tendency for people to excuse problems as being 'jokes' or 'no big deal'. My roommate wrote last year about having guys leer at her while taking pictures as she held hands with her girlfriend at SUDS last year. I see those guys every week at school. They laughed as they did it, and although I haven't confronted them about it by my friend's request, I can see them answering it now - that they didn't mean any harm, that it was no big deal. And I'm sure it wasn't a big deal to them, as straight white males. But it was to her. And this is to us.

Stereotypes are created and reinforced by things like this, small incidents that the perpetrators laugh at, and in particular, things that 'meant no harm'. You may not have intended it, but that doesn't stop the harm from happening, it just protects you from having to think about it. But a joke probably isn't that funny if you're the only one laughing.

It makes me sad to think that I immediately assumed that the vandal was an engineering student. Just like it makes me sad when people talk to me differently because they know I'm an engineer, especially when I know all the amazing things that go on within the engineering community, and all the amazingly welcoming people that are a part of it. However the careless actions that people have carried out because they were with their buddies having a drink and wanted a laugh, and because it was 'no big deal', have misrepresented us. We deserve to give ourselves better stereotypes than these.

When I was fitted for an engineering jacket, it seems I was also fitted with some stereotypes. I wish I could say I felt as proud wearing that jacket and those stereotypes as I do when I look at our mural.

So I urge all of you to remember that you are a part of a community, whether you accept it or not; whether you wear a jacket, belong to clubs, hang out in the Atrium, or not. And you are also a representative of that community. It's up to all of us to make the stereotypes belonging to engineers into ones we can be proud of.

The Greatest Asset in an Engineer's Tool belt: Knowledge in the Arts

Marissa Zhang
Writer

What comes to mind when we imagine the life of a typical "Artsie"? Someone who rolls out of bed at noon, barely goes to class, drinks on Thursday nights, and listens to music you've probably never heard of? With all of the homework and stress engineering students have to put up with, people can be quick to look down on arts students, and by extension, look down on the arts. Although it's very easy to feel smug about the opinion that engineering is one of the hardest undergraduate programs out there, knowledge in the arts is actually crucial for engineering success.

During the 19th century, higher education was based on the idea of *liberal education*. Liberal education is defined as education that provides students with transferable skills, and a stronger sense of ethics and values. One of the most important aspects of liberal education was that it encouraged students to think for themselves and challenged them to view the world with a critical eye. Derived from the teaching methods exercised in ancient Greece, liberal education was primarily done through the arts and the humanities.

The reason liberal education is so important is that it does more than cram a student's head with forgettable facts and knowledge; it encourages students to become more active citizens and actually care about what's happening around them. Students are affected every day by politics and law, and one day, many of the problems that the world is currently facing will be handed down to our generation. But how can we be expected to tackle these challenges when many of us are not politically or socially aware? While there are many students who do not need a course in politics to follow the news and participate in civic engagement, there remains a great number of students who don't even know how the Engineering Society government, much less the Canadian government, works.

In a society where technical degrees are considered more prestigious, we seem to have forgotten about the importance of a well-rounded education for university students. While liberal education and professional education seem to be divergent ideas, I would argue that knowledge in the arts and humanities could impact the success of an engineer a lot more than another math or science course. When engineers have knowledge in the arts and are well versed in subjects such as politics, law, and sociology, they go from being scientists and designers to leaders.

Civic engagement for engineers goes beyond simply following the news and taking the time to vote in every election. As engineers, we are designing systems and technology for the public. How can we be expected to design technology without even considering how it might interact with society? Human factors are extremely relevant to every single engineering discipline, and user experience has become equally important to technical design for new products. To fully understand the user, either psychologically or physically, it all comes back to the arts and humanities. Many of us will face tasks in our future roles that will not be able to be solved from a solely technical approach. Economic and political complexities that engineers must understand will stand in the way of the relevant problems. Educating an awareness into young engineers will help them to approach and solve these problems effectively.

Many schools in the United States have already incorporated arts into professional education. Columbia University, for instance, requires that all engineers complete the "Core Curriculum": a curriculum that includes courses in literature, art history, and economics, on top of a rigorous science and math workload. Cornell University has a similar setup for undergraduate engineers; engineers must take six courses from several liberal studies groups, such as cultural analysis and foreign languages.

I'm not suggesting that the key to being socially and politically aware is through an arts course, but I believe education in the arts is a good way to get students interested and involved with current events, and especially to help engineers better understand and become more interested with how technology interacts with society. While science and math provide students with the cold hard facts, arts and humanities teach students to be sceptical and think more about the big questions. STEM education is important, but it can lead students to be too narrowly educated. Education in the arts can make engineers into better thinkers, innovators, problem solvers, and leaders.

How can Rob Ford's actions be a lesson for all members of professional occupations?

Michael Woloszyn
Writer

What do lies, vulgarity, and crack have in common? None of them have a proper place in professional conduct. It comes as no surprise then, that Toronto Mayor Rob Ford's admission of smoking crack cocaine has sparked outcries amongst Torontonians.

Ford's admission to the use of crack cocaine comes alongside disturbing video of the mayor spewing obscenities during a drunken rant. Neither the use of crack cocaine or drunken rants are things you'd expect from the mayor of a large city. Worse is the fact that Ford denied allegations behind his use of crack cocaine for several months before finally admitting the truth. As a result, a strong majority of Torontonians feel that Rob Ford should step down from his position as mayor.

Toronto has been cheated out of mayor who can competently represent the city.

Don't get me wrong, Rob Ford is only human, and humans make mistakes. But, Ford's history of substance abuse is not something that you can just sweep under a rug. Sure, some may argue that Ford's ability to lead the City of Toronto has no correlation with the quality of his personal life. However, how can one possibly expect Ford to responsibly run Toronto if he can't even manage his own life? Toronto can do better.

The reality is that Ford's credibility as a politician has only been damaged by his reckless and cloddish behaviour. Ford's personal image is not in line with the conventional image expected from the mayor of Canada's largest city and as such has led to the Ford scandal. Part of the damage to Ford's profile stems from images and video of the mayor intoxicated. Thanks to the ever increasing access to technology, it has become easier than ever before to record just about anything at any time and anywhere. Because of this, the divide between professional and personal life is becoming increasingly thin. This has implications to members of all professional occupations, not just politicians. Be it doctors, lawyers, engineers, etc., all professionals must be increasingly careful with what aspects of their personal lives are exposed.

As engineering students, we're on the way to joining a prestigious occupation filled with many expectations and responsibilities. These additional responsibilities shouldn't come as a burden but instead be seen as society's positive belief that engineers have a strong ability to change the world. So take charge of your personal actions and know that with great power, comes great responsibility.

The First-Year Engi-New Years' Pledge

Eugene Hwang
Writer

Dana Umbrio
Writer

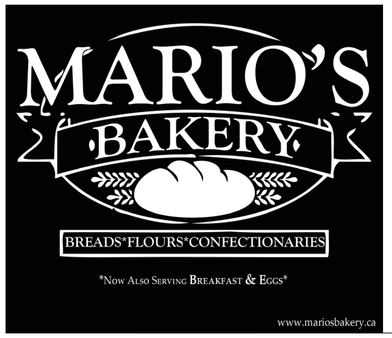
Now that 2014 is here, we can finally pick up the good habits we've been hoping to develop for the past year, and keep them for a good couple of days. Here are a few important tips I've received from professors, upper years, and parents throughout my first semester in Engineering.

1. Watch your diet

Contain your urge to stress eat everything in sight. In addition to the well-known Freshman 15 effect, diet can really determine whether you get a good hour of studying done or pass out from a food coma. Overstuffing yourself leads to a nice long nap, that tends to become actual sleep. Plan out what to eat at crucial times, such as right before an exam. Bananas are known to be good brain food, and as proven by several of us Frosh this year, taking 3 shots before an ESP exam might not be as bad an idea as it sounds.

2. Don't take up too much on your plate and manage your time

Although it's customary for us energetic youngsters to speedily build up impressive resumes, don't spread every square inch of your abilities across all possible responsibilities. We may not realize yet, but our bodies do have thresholds, and when said thresholds are crossed, there may be no going back. Balance an above average academic life with some extra-curriculars and an enjoyable social life. So in summary, don't take Eng Sci.



3. Get involved outside of class

That being said, avoid hermit status at all costs! Being involved in class really helps avoid insanity in times of finals chaos. Granted, a season of "Suits" should have a similar stress relieving effect, but only one gives you a one up at a job interview. As well, joining extra-curriculars gives you an opportunity to make yet another circle of friends that you can have fun with and rely on, or just to go to when you get sick of your other friends.

4. Try to make some friends

It's important to remember than almost everyone in first year, not just you, is new to university. Even something as simple as a quick introduction or chat in class will help lead to good study groups and good times.

5. Get moving

Exercise seems to be a heavily avoided concept by many in our modern North American society. But not only will it reduce the dreaded Freshman 15, it can also give us a mental boost. A good run in the park or swim at Hart House pool will get your blood pumping and your brain juices flowing, and an hour long study session interrupted by Facebook and naps can become a 5 minute focussed session. This notion is supported by the beloved Professor Rose of Computer Fundamentals.

6. Ask for help if needed

Don't feel bad if you're stuck on something or don't understand a concept, as most of the time, someone else doesn't either. Do not hesitate to ask for help from your peers, as well as upper year students, TA's, and your professors. Take advantage of the First Year Office's numerous resources, such as academic and success counseling, facilitated study groups, and tips for good study habits.

7. Form study groups

Putting different minds together allows new ideas to flow, and can really help in understanding a new subject. Plus, discussing a concept in a small group of your peers makes it a lot easier to understand and gives more opportunity to ask questions and clear up confusing concepts – especially compared to learning it from a prof who is also teaching a hundred other students.

8. Don't leave your work to the last minute

This is possibly the most important skill that one can develop. There is a huge difference between completing all work to the best of your ability well before the deadline and rushing to finish half of it at 2 AM the night before it's due. Avoid unnecessary extreme stress and give yourself more time to think and ask questions.

9. Find a good place to study where you can focus the best

For me (Dana), the library is the best place to study and focus, instead of at home where I can get easily distracted by the TV and my computer. If the library bores you, try switching libraries, or library study rooms. You could try some of the empty classrooms and study areas in Bahen as well.

10. Enjoy the weekend

The weekend is definitely a time to catch up on what you missed during the week, but it is also a time to relax and socialize. Take a Friday night to catch up with friends and give your mind a break. I (Dana) find that it's easier to get things done on Saturday if I relax Friday night.

Why Do Students Have To Abhor School?

VISMIT JOSHI
CONTRIBUTOR

Remember, challenge does not mean stress. Stress is a poison that can ruin a life.

More than feeling challenged, many students feel restricted, jailed, and unhappy with life in this education system.

Isn't education meant to be an exhilarating path to enlightenment and wisdom? The pressure of exams, a restricted curriculum, and a rigidly unattractive style of teaching take away from the spirit of free learning and disguise the beauty of the ideas being conveyed. Although no one seems to care about changing this due to a seemingly unknown reason, it doesn't have to be this way.

Someone who disagrees with this idea may think that students are taught and evaluated in this manner because there is no other numerical means of gauging a student's success. Yes, they are right! There is no other way… no easier way. They may also think that the education system is good as it is because people have been passing through it for hundreds of years; it works. That is true. It is very easy to just pass through university because all you need to do is to recognize that university life is a game and make sure you play it well.

So, a majority of us can survive, and when we grow up, we do nothing to change it because it's over for us. Who is concerned anymore? Therefore, a system that is survivable is not necessarily effective. So things need to change. In fact, optimal methods of learning that are much more effective in boosting student interest and success already exist. All that needs to be done is to put some effort into (majorly) integrating them into our education system, with emphasis on the post-secondary level.

Alright, this change is not meant for every student, but for only the few who actually have the spirit to learn and do something with their knowledge. The reason nothing has changed and all geniuses have had a history of struggling with life in this society is because there is only a small fraction of us that fit that category. Since majority rules, that fraction is marginalized and geniuses always have to struggle. It doesn't have to be like that. After all, these are the people who will improve the lives of the "majority group" of people in the future, so why should they be mistreated?

Now, take a moment to think about all the great minds we have had in the past. Some of the names that may have popped up could have been: Newton, Einstein, Tesla, John Nash (portrayed in the movie "A Beautiful Mind"), Abraham Lincoln, Beethoven, Charles Darwin, Charles Dickens, Michelangelo, and the list goes on. What is in common between all these personalities? The fact is, all of them were unusual, weird, and abnormal. It is a scientifically proven fact that intelligence is linked to mental illness.

For example, Einstein was thrown out of school and rejected by society as a mentally ill outcast. He did succeed, although his chances were extremely low given the situation. So, he simply got lucky. A lot of his counterparts may have had the same luck because they were rich and could afford to put themselves out there (i.e. Darwin). What happened to the rest of the creative minds of the past that were not recognized? They died with their beautiful, creative minds living at the bottom of the barrel. What this suggests is that the mentally disabled (i.e. students with ADD, Anxiety, Autism, and OCD – for whom the current education system is not meant) will never shine unless they get lucky. Thus, we should give more importance to those types of students and give them the opportunity to receive as much merit as the other students.

At university, the main goal of the student is to be mentally and practically ready for the industry by achieving something called a "degree". I believe there is much more to a degree than grades achieved on pen-and-paper exams. There exists skill, creativity, management, personality, passion for the subject area, and a bold uniqueness to the person that possesses his or her degree. Unfortunately, these aspects of a student are not taken into consideration when a student's achievement is gauged.

Even worse, many students with the highest potential to contribute to our world are left behind by those who are good at writing exams—but not necessarily readier for the industry. It is also likely that those exam writing experts who can solve problems on paper may simply lack the essential ingenuity to do something with their knowledge in real life.

What is the problem, then? The culprits are: teaching style, testing style, exam anxiety, academic pressure, lack of personalized training, and areas of a student's mind—or lack thereof—which are tested. And there is a big one that applies to heavy programs like engineering: dense schedules that put students under information overload in a tight schedule to prevent them from actually unleashing their creativity by giving them time to mull over what they just learned. Doesn't this sound like a manufacturing factory? Education is being mass produced! This is bad.

Furthermore, every person has their own learning style, but university only caters to one of them (which is known to be incompatible with people who can really "do things" in life) because of its monotonous instruction style. There may be more competent students who are failing at the system just because they don't learn that way and they are not good at playing this game. And our system has been like this for centuries, which means that we have likely lost many Einsteins in the crowd because the system did not accommodate their style of learning. Potential geniuses should be identified and catered for, just as we nurture a seed so it can become a fruitful tree. Therefore, it is very logical that the system's way should not determine the success, self-esteem, and competency of a student, as I have observed that this is grossly ineffective.

Now, it is clear that something urgently needs to be done to revolutionize the system (for those who need it).

You may be wondering: what will this optimal education system look like? The fact is, it will take a lot of effort to change the current state of education. Some institutions may refuse to change because they are a business and the capital investment will harm their profits. Someone might say that it is too early to start, but if we don't start now, when will we? A few years from now the same excuse will be used because we didn't start earlier. In the end, the benefits of this transformation will outweigh the costs greatly. As for our vision, this new type of school will encourage collaboration more than competition, there will be ongoing observation to determine the success of the students, learning will be personalized, and students will have extra time to think beyond the curriculum and make creative connections with what they learn. This explorative approach to learning will definitely be more challenging than the current method (remember, challenge does not mean stress), but students will truly enjoy it and actually value what they learned for the rest of their lives.

To conclude, let's talk about how education fits into the big picture of a human life. All of us are educated from the age of five through the age of 23 and beyond. Once most people hit their thirties, life becomes too regular to be fun. Retirement comes just around the corner, diseases appear, and there comes death. In society, it is widely accepted that one should enjoy life in youth. If the age when a person is educated is the golden time of a person's life when learning, development, and enjoyment happens, then why should it be ruined by institutional stress? We want to be happy in the most important part of our lives, so we will make sure that our future generation gets that happiness by doing something about the issue at hand.

The system must be improved. "No one can do everything, but everyone can do something." – Unknown

What will you do?

Author's note: These ideas are strictly opinionated, if your opinion is the complete opposite, that is fine, too. Hope you enjoyed reading.

The Secret World of Library Jargon

TRACY ZAHRADNIK
UOFT ENGINEERING LIBRARY STAFF

It is no secret that people love professional jargon. Sometimes at the library, when students ask for research help librarians are left scratching their heads! Just what is meant by **froth floatation** (part of a latté?), **escape sequence** (how to get away from a bad blind date?), **acid attack** (Pepto-Bismol to the rescue?), or **zero point** (a bad midterm mark?)?

We, at the library, are no different! Often we let slip some of our secret library jargon when talking to students (or Profs!) - leaving them dazed and confused. To help demystify the (what we believe to be) exciting world of libraries, we have decided to share some secret library jargon with you…

Facets or faceted search: No, these are not kitchen water taps but a fandangle way of saying the 'characteristics' of an item. In the library catalogue, we describe the different facets of library items such as location (Online, Engineering & Computer Science Library, and Aerospace) or subject (aeronautics, data processing, and product design). These facets can be used to narrow or filter a search to find items you need.

Controlled vocabulary (aka ontology): Did your mother ever tell you to watch your P's and Q's or she would wash your mouth out with soap? That is different from the controlled vocabulary we use in libraries. To us, "controlled vocabulary" is a standardized terminology used to index things such as journal articles in databases. Using controlled vocabulary in databases (like the "Thesaurus Search" function in Compendex) can help to increase the accuracy in which you can find articles.

Interlibrary loan: Interlibrary loan, often referred by the acronym ILL, has nothing to do with being sick. It means borrowing a book or getting a journal article from a non-University of Toronto Library. As a further point to confuse you, the system used to coordinate interlibrary loans is called RACER. You can use the RACER website to create an account and fill out an ILL request form.

Stacks: At first you might think, hey neato, engineers and librarian share some jargon! However, while engineers think of stacks as a type of C++ container adapter, librarians refer to an area where books are stored as stacks. Specifically, at the Engineering & Computer Science Library when we say, "That book is located in the stacks", we mean the book is on the 2nd or 3rd floor of the library. We also have "periodical stacks", on the main floor where we keep journals.

Science and the World Wars

ISRAEL GONZALEZ
WRITER

As the 100th anniversary of the assassination of Franz Ferdinand and World War One approaches, I wanted to take the time to meditate on the effects of both world wars on science.

This article will examine a number of scientists and their war stories.

Part 1: World War One

Science was at the forefront of the Great War in more ways than one. Some scientists gave their lives for their countries, as did Henry Moseley, a British chemist killed at Gallipoli in 1915. Moseley had profound effects in how we study chemistry today through his work on getting better values for atomic numbers of elements. In France, Marie Curie's work on X-ray technology saved countless lives and limbs during the war. She helped equip ambulances with X-rays during the war, making diagnosis and treatment much easier.

On the other side of the war, the most prominent scientific figure was that of Fritz Haber. Like Curie and Moseley, Haber was a chemist whose work came in handy during the war. Most students of recent history are aware of the usage of poison gas during the war by the Germans and later on the allies. Haber was employed as a consultant during the war where he was instrumental in the German war strategy regarding the usage of poison gas. He often was consulted for the organization of poison gas attacks and defense against them. After the war he would serve on a League of Nations committee dealing with chemical warfare.

Part 2: Between the Wars

Between the wars, some scientists became conspicuously involved in political movements. The issuing of the Balfour Declaration spurred scientists of Jewish heritage into activism for the cause of the Jewish homeland. Albert Einstein became a very visible figure in the struggle for a Jewish state and was often seen with chemist and activist Chaim Weizmann. Weizmann would prove instrumental in building the state of Israel and became its first president upon independence in 1948. Einstein later declined becoming president himself, but featured in Israeli currency later on.

In the USA, the 20's were a time of economic and academic boom. Philo Farnsworth conceived the idea of television, changing the course of human history; Einstein would work on the Manhattan project with Robert Oppenheimer; and Pluto was discovered in 1930. Amidst the jazz, America was rising not only as an economic and military power, but also as an academic power, and that last trait would turn America into the academic capital of the world with a plethora of renowned institutes of higher learning. In mathematics, Jesse Douglas of MIT was among the first Fields Medal winners. The Fields Medal of course was a University of Toronto brainchild as it was named after J.C. Fields, a Canadian mathematician.

Part 3: World War Two

Hitler's rise to power touched off an exodus of scientists and other thinkers from Germany. Haber went to the UK and Einstein went to the USA, where he was to become America's and the world's first celebrity scientist and one of the most marketable names and faces in history. In the years that followed the world went back to war and a deadly competition was about to reach its explosive climax. America, Japan and Germany were competing to develop the atomic bomb. In Germany, the project was spearheaded by Nobel Prize winner Werner Heisenberg. Uncertain of who he is? Well, he was appropriately enough the father of the Uncertainty Principle and lived and worked in very uncertain times. He was unable to beat Einstein, Oppenheimer and company to the bomb and was eventually taken prisoner by the Americans. He was captured in part due to the efforts of Moe Berg, a US spy and baseball's ultimate polyglot who had retired from the sport a few years before the war.

The UK during World War Two employed science greatly and the result was one of the great stories in science. The father of computer science, Alan Turing was also instrumental to the war effort because it was through his efforts and those of fellow codebreakers at Bletchley Park that the German enigma machine was at last deciphered and important information on the Germans divulged. The British Bombe was essentially an early computer developed by Turing as part of the war effort. The life of Turing is fascinating because it sheds great light on the role of science and mathematics during World War Two. His work made him Britain's answer to Einstein, a celebrity scientist who has captured the imaginations of generations of children in the UK and abroad. The only difference is that Turing died young and never really lived to see Britain appreciate his work like the USA did Einstein's during his lifetime.

Part 4: After the Wars

Canada would prove to be an important beneficiary of Germans and other Europeans fleeing Hitler. Gerhard Herzberg would win the Nobel Prize in chemistry in 1971, giving Canada its first Nobel Prize in science. (Lester Pearson had won the peace prize previously.) Einstein became a committed pacifist and Oppenheimer would later see himself as "death, the destroyer of worlds". Heisenberg was released after the war and continued to research and teach.

The experiences of these men and women would play profound roles in shaping the world in which we live today. The work of Curie made women more prominent in science than ever before. Moseley became a martyr for the Union Jack and a hero to Britons. The partnership between Weizmann and Einstein resulted in the birth of Israel (not me, the country of course), itself now a technological and scientific powerhouse. Finally, the work of Haber profoundly changed chemistry and his work lives on in agriculture and other related fields. His work also gave him a Nobel Prize in Chemistry, one of many going to German scientists during the first decades of the 20th century.

The world wars turned scientists into soldiers, humanitarians, political leaders, nation builders and celebrities among other things. The face of war would change forever within those 30 years and so would the roles and job descriptions of those who fight them. Hence, you have one more thing to ponder as you remember the world wars and their unique effect on humanity.

References and useful resources associated with this article can be found at cannon.skule.ca

Science & Technology in 2014: The Top 10

We saw some amazing new technology in 2013. What will 2014 bring?

CLARA LLOYD
SECTION EDITOR - SCIENCE & TECHNOLOGY

1 Lasers send data through space

January 2013 saw the first laser transmission of data through space: an image of the Mona Lisa was received by the Lunar Reconnaissance Orbiter. Other tests indicate that this method of transmitting data is indeed practical. 2014 should see a large increase in the amount of data that flows to and from space, and will change the way humans explore the solar system. Rovers would be able to send back more detailed data … and astronauts could even Skype back home!

2 The Human Brain Project

In October 2013, The Human Brain Project began at the Swiss Federal Institute of Technology in Lausanne. This project aims to create the first complete computer simulation of the human brain. Other similar projects will be undertaken in 2014, led by the U.S. government's Brain Initiative. Even though these projects will take up to a decade, researchers have just developed the technology that makes the modelling of billions of individual neurons possible—and 2014 is the year in which many of these projects can start.

3 Dismantling chemical weapons

The U.S. Army plans to construct and use Field Deployable Hydrolysis Systems (FDHS). FDHS's are essentially mobile decontamination labs that can rapidly neutralize 5 to 25 tonnes of dangerous chemical-warfare agents each day. The FDHS technology could be used in countries, such as Syria, that do not currently have the facilities to destroy their chemical weapons supplies.

4 Drone use will be regulated

The Federal Aviation Administration (FAA) will issue a draft rule that regulates the use of drones (under 55 lbs) in U.S. airspace. Many flight tests planned for 2014 will determine the future of these unmanned aircraft. In fact, the FAA has already cleared many applicants to fly in a not-for-profit capacity.

5 Celebrities go to space

Katy Perry, Lady Gaga, and Justin Bieber have apparently booked seats aboard the Virgin Galactic spaceship, the first commercial spacecraft to take paying passengers into space. Need I say more?

6 Creating walls underground

Last August it was discovered that a damaged tank at the Fukushima nuclear power plant was leaking contaminated water into the ground. In 2014, the Tokyo Power Electric Company proposes creating an underground wall made of ice in order to prevent the groundwater from entering ocean waters. A piping system will be installed around the power plant and chilled coolant (calcium chloride brine) will be pumped through the pipes, circulating continuously and freezing the ground. This frozen ground forms an underground wall of ice, blocking water flow.

7 Less invasive cancer diagnoses

Identifying cancer is usually only possible after the symptoms have developed. Exosomes are tiny sacs that travel through fluids in the body (blood, urine, saliva), transporting genetic material and proteins between cells. The proteins or genetic material carried by these exosomes can be used as an indicator for the presence of potentially malignant mutations. Two companies are competing for the 2014 release of the first commercial exosome test, useful for prostate and breast cancers. Research is under way for extending the application of similar exosome tests to other cancers.

8 Renewable Offshore Energy…?

Construction of the first offshore wind farm in the U.S. has been planned for 2014. The construction of such wind farms opens up the possibility for increased use of renewable wind energy in the U.S.; indeed, plans are already under way for wind farms off Massachusetts and Rhode Island.

9 Physicists create spy-proof code

As if quantum physics wasn't confusing enough, Quantum Key Distribution (QKD) is, in theory, an unbreakable encryption protocol using the concept that observing information changes it. A QKD system uses a randomly generated key encoded on light particles, shared on fiber-optic cables, and then used to encrypt sensitive data. If the data is intercepted or detected, the photons will be altered, indicating that a new key is necessary. Although fiber-optic cables are the current mode of photon transmittal, physicists at the Institute of Quantum Computing in Waterloo, Ontario, are developing microsatellites that could send encoded photons across the globe in a beam. The first commercial quantum-encrypted data network is planned for release in San Diego by September 2014.

10 Infectious diseases re-emerge

Although great advances have been made in the area of infectious diseases over the last century, some illnesses have begun to reappear, for various reasons. Increased travel and global food trade means diseases spread in a wider range across the globe than ever before. In other cases, parents fail to vaccinate their children against preventable diseases in a timely manner, and in other cases, poor antibiotic management has led to an expected re-emergence of infectious diseases in 2014.

References and useful resources associated with this article can be found at cannon.skule.ca

Intimacy on the Subway

LOBNA EL-GAMMAL
SECTION EDITOR - ARTS & CULTURE

The silver pole bears the juxtaposition of our palms and their colors so eloquently that for a moment I think we are the same color. Yet a second glance at my skin, inches away from his, flashes images of dalmatians in my head. He is dark and I am light. He is plenty with melanoma, and I am poor of it. He is strong in resisting the sun, and I am weak.

This metal pole houses many palms a day and is nourished by the sweat and stories of subway riders. Its occupancy level fluctuates with time, peaking twice a day, once in the early morning and once in the afternoon. I am but 5 fingers amongst millions that comprise the second peak of today. His hand contributes another 5 fingers.

I wonder what his hand grips beyond the subway doors. Is he a writer?

A teacher?

A construction worker?

A scientist?

A florist?

A painter?

A chef?

An athlete?

Does this hand grip pens that birth stories or chalk whose powder creates traces of knowledge? Maybe his hand is accustomed to the grip of tools or that of latex gloves. Do his hands feel at ease with the prick of rose thorns or dried paint brush tips? His hand may be trained to feel the beating of a human heart or to feel the hollowness of a watermelon, or a basketball. Better yet, his hand may not always pursue a single talent, but remain open, palms ready to accept.

Crowded subways, like packed concerts, are prime locations for intimacy, both appropriate and inappropriate. The subway jerks to a stop and I pray softly this isn't his destination. His fingers relax, but not enough to indicate his departure. I am blessed with a peek into his inner palm, rosy in complexion and furrowed in texture.

I imagine the fingers his hands have nestled themselves in between. A lover with whom he exchanges kisses and secrets. A daughter whom he is afraid to let cross the streets on her own. A grandfather whom he begged to stay in this life as long as possible, disregarding the angel of death. Maybe his fingers nestle themselves within one another every night as their embrace lays on his forehead, concealing the tears born of a heartfelt prayer to the Lord.

The subway jerks again, but I am not as fortunate this time. His hand departs the pole quickly and all that's left for me to contemplate is his imprint on the shiny pole, short lived as it may be. It will either fade into nothingness soon or be buried under the grip of another hand. The intimacy of subway rides is fake, and I wish I could put a name to the dry wrinkles across his knuckles.

Michael

MEGHAN LELE
WRITER

"It's good to see you! Come in, have a seat, and make yourself comfortable," said Dr. Friedlander, holding the door open for his next patient. A young man of eighteen years walked in and seated himself on the couch opposite the doctor's chair. He sported black and white formal attire from head to toe, and wore a solemn look of dissatisfaction.

"It's been a long time, hasn't it? How have you been these past few months, Michael?" inquired the doctor. "I see you still like to dress to impress."

"I've been okay. Things could be better, but they are what they are," replied the young man, unwilling to reciprocate the warmth of Dr. Friedlander.

"How did school go? I would be surprised if exams were very challenging for someone like you."

"It was okay. They were okay." Michael sat there with his arms crossed, staring contemplatively at a decorative sculpture on the table.

"Based on how first semester went for you, you seem to be at the top of your class. Additionally, unlike most of your peers, you were able to secure a summer internship related to your field of study. Things seem to be going your way, Michael; why are things always 'okay'? Do you feel dissatisfied with your success -- at least, that's what I'd call it -- and if so, why?" probed the doctor, noting that Michael had not changed very much since their last meeting despite having a job for the first time. Dr. Friedlander had expected that Michael's time (albeit short) in the workforce would have alerted him to the plight and hardships experienced by others in the world, and would have enabled him to be happier and more appreciative of the current state of his affairs.

"Anyone in my position could have done what I have done, Doc. I have two arms, two legs, two eyes, two ears, just like everyone else. It's just hard for me to be excited about something that's not a big deal," replied the stoic patient.

"I disagree, but let's say, for the purposes of discussion, that I accept your premise. If what you say is true, why then have so few of your peers been able to accomplish the same things?"

"They don't want to. They're just not trying hard enough," said Michael while shaking his head, seemingly dismissively at his colleagues.

Though years of experience allowed him to hide it, Dr. Friedlander was infuriated by his patient's assertion.

"Let me assure you that it is not that simple, Michael. There are members of your class who have and continue to invest their blood, sweat and tears into realizing their goals, but have not been able to make them happen. I know you dislike comparing yourself to your immediate peers, but I just want to make you aware of this fact. When you spend time around your peers, do you get the impression that they don't want to succeed?" asked the doctor.

"Well, Doc, I don't spend that much time around my peers or friends. The reason being is that, based on my experience, all they do is complain about how hard university life is academically instead of working hard to alleviate that burden. They're so comfortable with being mediocre that it disgusts me. Note that it is the lack of effort that disgusts me; I respect those who try their hardest to finish every task at hand to the best of their ability, even if the results aren't what they expected or wanted. But believe me when I say that most of my peers aren't even close to exerting themselves," replied Michael.

"Let's focus on your expectations. If you are unhappy and dissatisfied, there must be some goals or dreams that you hoped you would have accomplished by this point in time. Could you tell me a bit about that?" inquired Dr. Friedlander, hoping to reposition the focus of the conversation on his patient and uncover his motivations.

"Not really. I'm just an average Joe doing things an average Joe would do. It's not like I've cured cancer or anything…"

"You mentioned in the past that your father has been the breadwinner in the family for 30 years now, and that your mother won the battle against leukemia a year ago. Additionally, at the time of our last meeting, your brother was struggling to find employment despite being a fresh computer science graduate. You expressed a great desire to help your family back then, and I could see that it motivated you to try and excel academically and professionally. You're in second year now, and you are approaching a point in your life where you can act on that desire, and realistically alleviate the burden on your father's shoulders. Do you have any plans for that?"

"Yeah, as a matter of fact, I do. Plans are subject to change, obviously, but as far as I know right now, I'm going to work in the summers between school years to pay for my degree, and right after graduation, I'm heading to a tax haven to mint as much money as I can while I'm young. Preferably somewhere in the Middle East, but Washington state is the backup. My father is going to be able to retire in his fifties if he wants to, or he can keep working for fun. My mother can rest and recover in peace without having to worry about the cost of her prescriptions or any additional treatment. My brother … maybe the humiliation will spur him on to look harder for a job." said Michael, smiling as he stared into the clear blue sky through the office windows. Dr. Friedlander couldn't help but smile as well; this was the first time that he had seen Michael display emotion and show great compassion for others.

"Michael, I think that's a very selfless and noble ambition. A lot of other men and women your age don't think of giving back to their families that quickly and to that degree. In my opinion, this is a very potent, healthy and positive motivator for you, and that's really good to see. While I'm sure your parents will appreciate this, you'll always be their son, and they'll want you to take good care of yourself as well. Restarting your life on another continent can be a very arduous ordeal; have you given any thought to how you will take care of yourself in this plan of yours?"

"Going to the other side of the planet is the best part of this whole idea! I hate going to school, and it is the primary source of discontent in my life. The farther I am from school -- and that includes the people at school -- the happier I am. And as for taking care of myself, they have food, water, and air in all of my prospective destinations. I do not lead a life of luxury, so comfortable accommodations shouldn't be difficult to find. All I really need on top of that is an internet connection and a tennis court and I'm set, Doc."

"What about your friends at school, Michael?"

"Like I said, can't wait to get away. And besides, my best friend will be there wherever I end up going, even if it's not any of the places I have in mind right now."

"I don't understand what you mean."

"His name is Benjamin Franklin," said Michael, with a sly grin on his face.

"Michael, it's wonderful that you care so deeply for your family and want to take care of them, but human beings are social creatures. Money can't address your need for meaningful social interaction with other human beings. How will you be happy and sane with your family on the other side of the planet and no friends to … 'hang out' with?"

"Don't be a cliche, Doc. Money can buy happiness, and you know it."

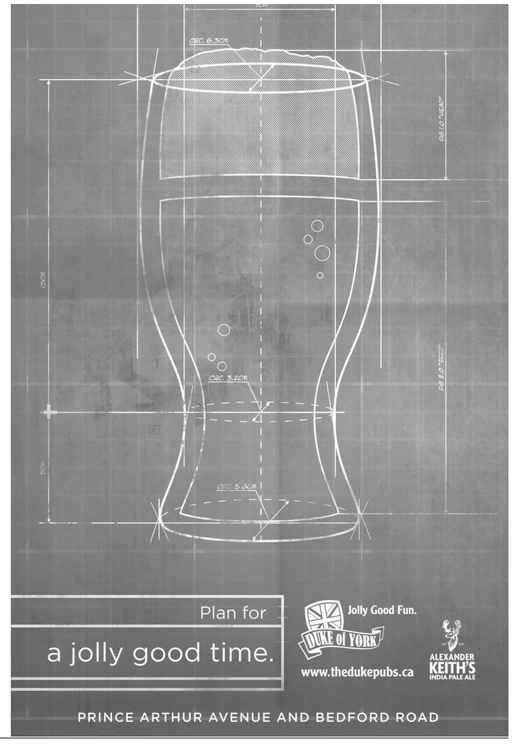
"I don't, Michael. Why do you say that?"

"Once you earn some money, it's yours until you let it go. Provided you have enough, you can purchase whatever comforts and luxuries you want. Given that I don't plan to start a family, I'm certain I'll have enough."

"What about the company of friends? Of other people? You don't think that's a prerequisite for happiness?"

Michael looked at the clock on the wall behind the doctor, and picked himself up after realizing that their time together for the day was almost at an end. He stretched for a few seconds, jutting his arms out wide, before removing three twenty dollar bills from his pocket and placing them on the table. He walked toward the door of the office and opened it before turning back to look at the doctor.

"The thing I grew to hate about people is how their loyalty can change in the blink of an eye. That's where money is different. Money doesn't judge you, doctor," said Michael before leaving Dr. Friedlander, whose gaze was fixed on the face of Queen Elizabeth II.



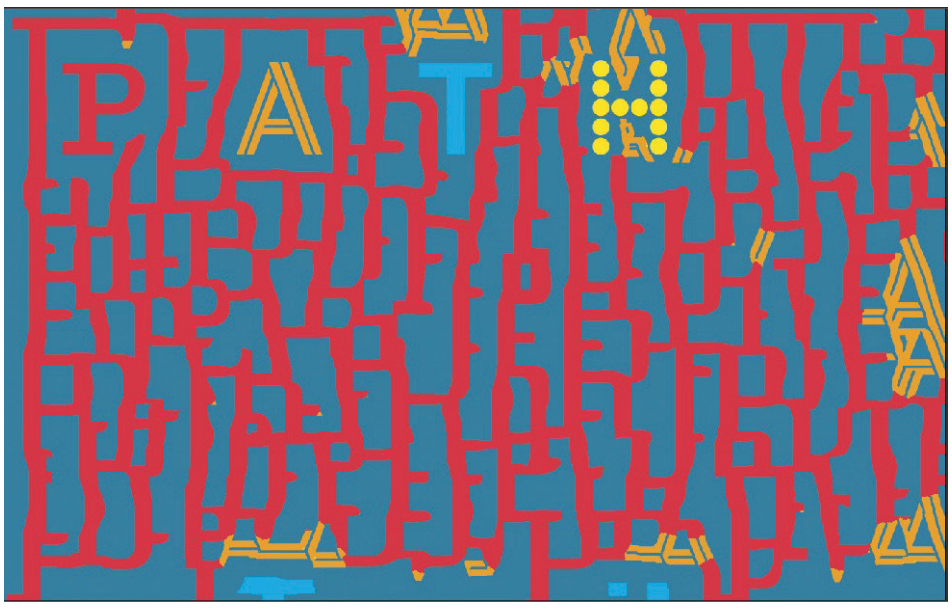
Algorythms

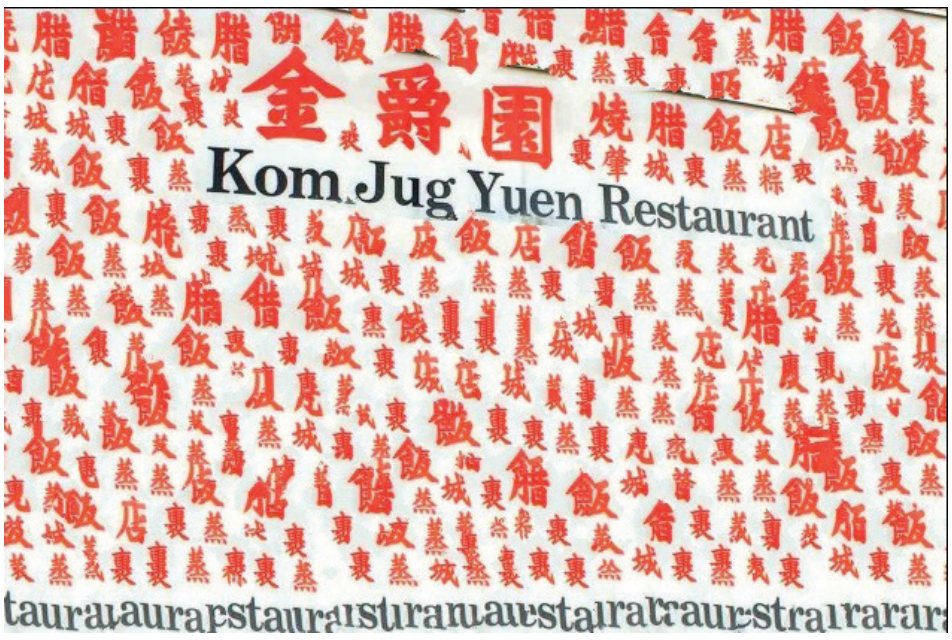
NIKITA DAWE
CONTRIBUTOR

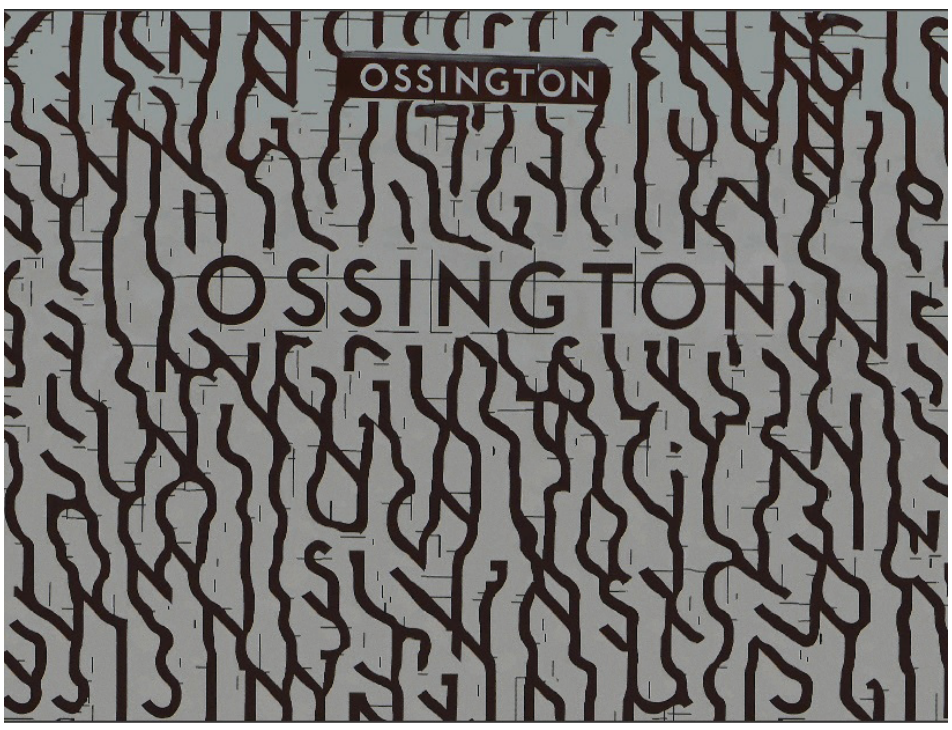
Experimental poet Christian Bok calls it visual poetry. Designer and curator Silvio Lorusso calls it content-aware typography. It started as a mistake – Photoshop has a feature called Content Aware which is designed to "make objects magically disappear" when editing a photo. If you abuse the feature and try to fill the majority of an image while leaving some text component visible, the algorithm generates words, rhythmic repetitions, and imagery in positive and negative space.

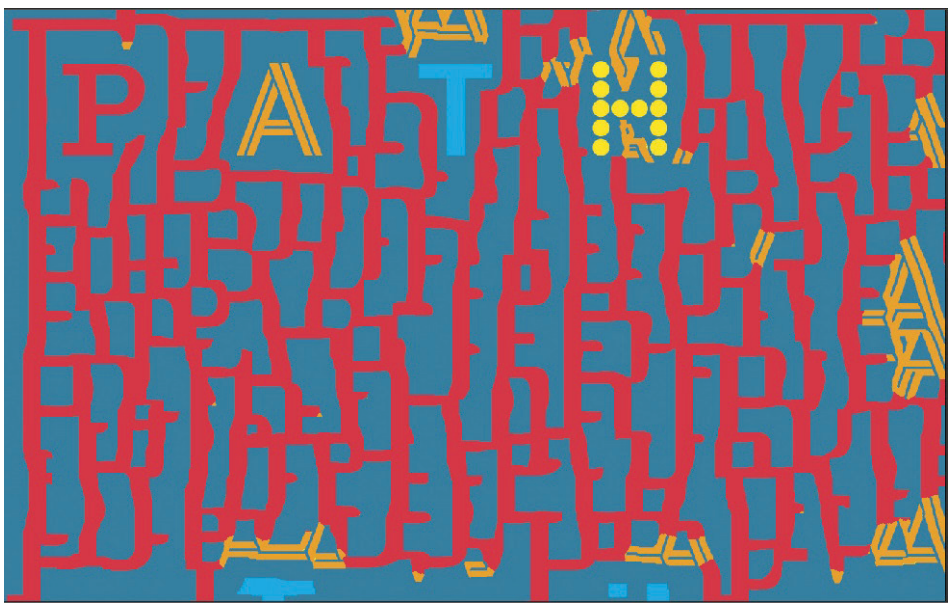
I began with photographs that define Toronto to me, as well as some engineering-related images. They are all stolen, so thank you various photographers and legal entities. I used Inpaint instead of Photoshop. This is a small selection of the output:

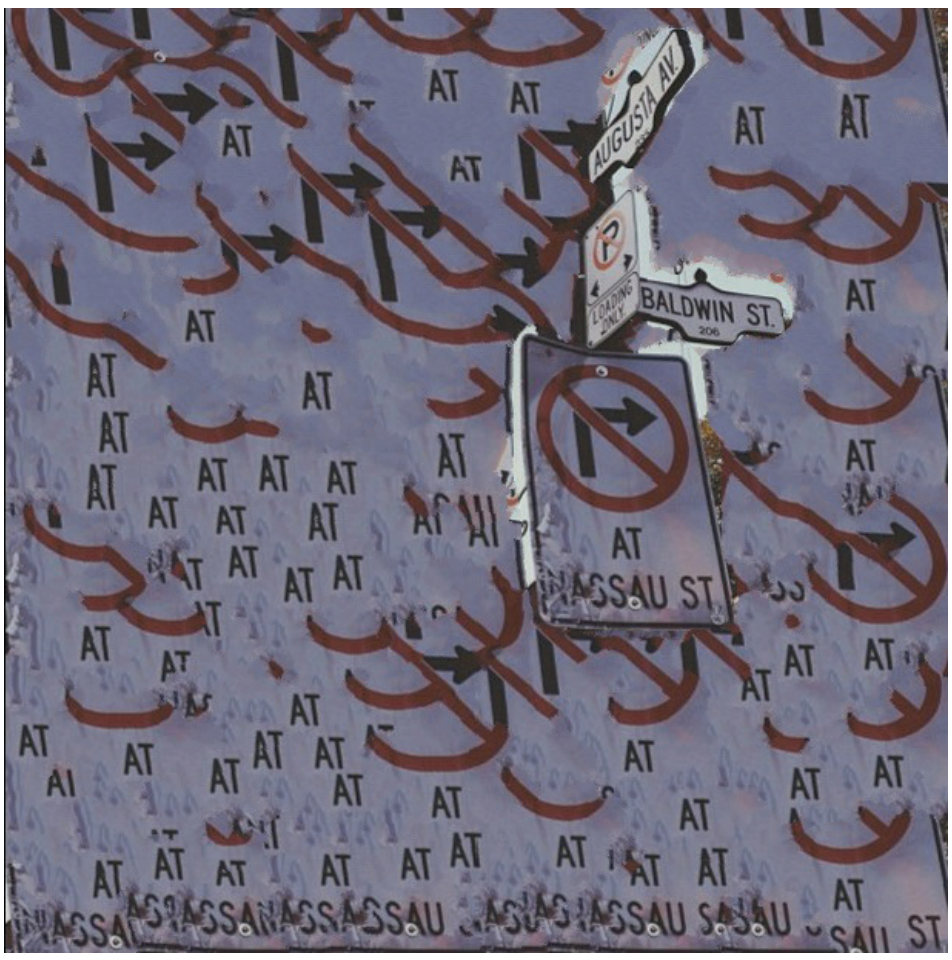
References and many more of Nikita's images can be found at cannon.skule.ca











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